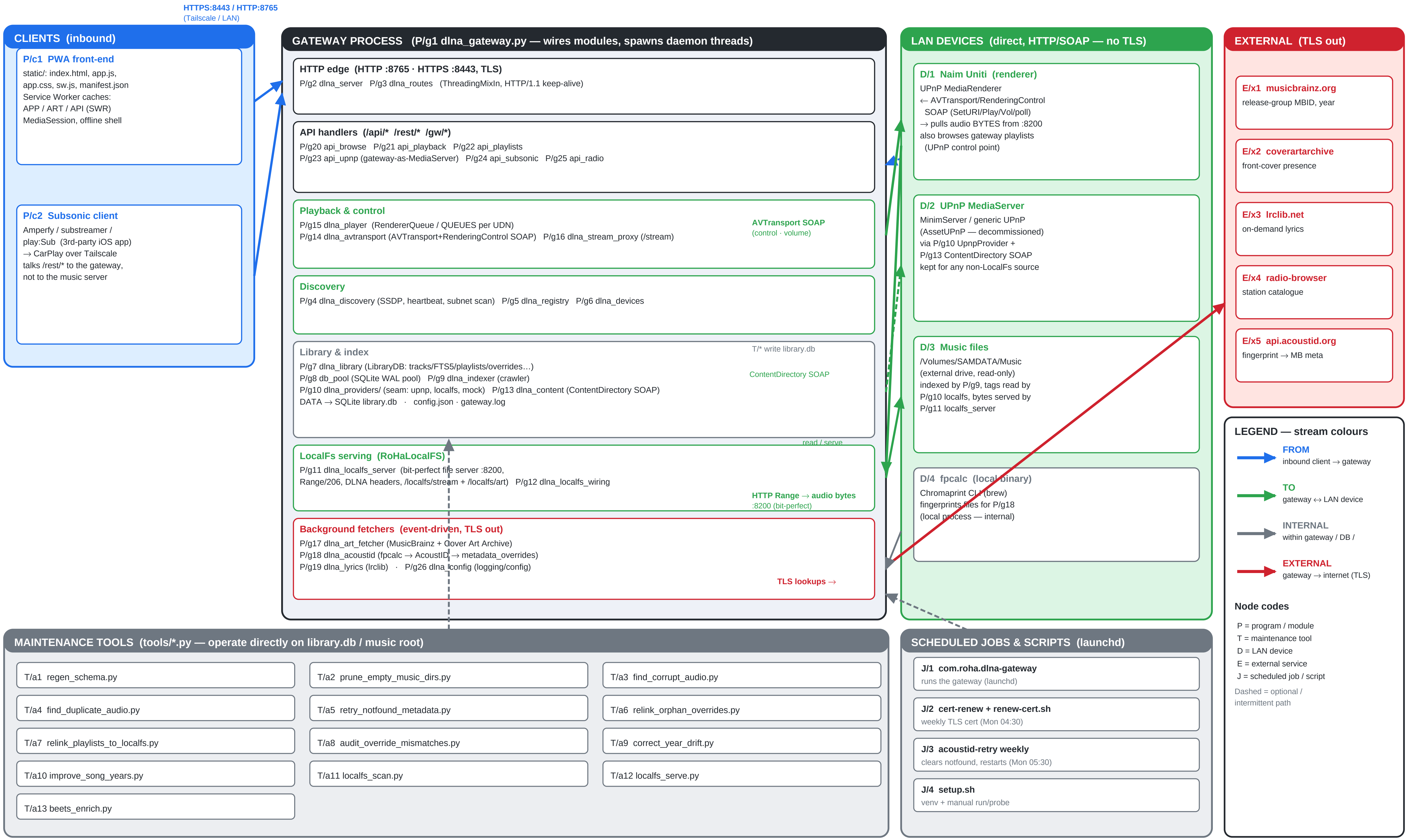
DLNA Gateway — Current Architecture (1.x)

What runs where · stream colours show direction/scope · node codes (P/T/D/E/J) index into the lists overleaf



Node lists — every box in the drawing

Codes match the diagram. **P** = program / module, **T** = tool, **D** = device, **E** = external service, **J** = scheduled job.

Application programs & modules (P/*)

Code	File / location	Responsibility
P/c1	static/index.html, app.js, app.css, sw.js, manifest.json	PWA front-end. Browse/playback UI, MediaSession lock-screen, Service-Worker caches (APP shell SWR, ART cache-first, API stale-while-revalidate), offline shell.
P/c2	(3rd-party iOS app — Amperfy / substreamer / play:Sub)	Subsonic client for CarPlay over Tailscale. Talks /rest/* to the gateway; never to the music server directly.
P/g1	dlna_gateway.py	Main entry. Wires every module; spawns daemon threads (SSDP listener, pre-prober, subnet scanner, heartbeat, gateway announcer, HTTP + optional HTTPS servers).
P/g2	dlna_server.py	Threaded HTTP/HTTPS server (BaseHTTPRequestHandler + ThreadingMixIn). HTTP/1.1 keep-alive + 15 s idle timeout. Delegates routing to P/g3.
P/g3	dlna_routes.py	GET_ROUTES / POST_ROUTES path→handler maps.
P/g4	dlna_discovery.py	SSDP multicast listener, probe, subnet scanner, renderer/server heartbeat. Holds SERVERS / RENDERERS singletons.
P/g5	dlna_registry.py	Data classes + thread-safe ServerRegistry / RendererRegistry.
P/g6	dlna_devices.py	DeviceRoleCache — in-memory mirror of device_roles for zero-latency classification.
P/g7	dlna_library.py	LibraryDB — SQLite index, FTS5 search, playlists, album_art, play_counts, lyrics, metadata_overrides, favourites, radio. Composition root for DB-owning singletons + fetchers.
P/g8	db_pool.py	SQLite connection pool — WAL, thread-local conns, write serialization.
P/g9	dlna_indexer.py	Background crawler that walks a provider and populates LibraryDB (with FTS self-heal).
P/g10	dlna_providers/ (__init__, upnp.py, localfs.py, mock.py)	LibraryProvider seam + registry. UpnpProvider wraps the UPnP SOAP path (kept for MinimServer); LocalFsProvider is the in-process backend (mutagen, content-hashed ids, watchdog).
P/g11	dlna_localfs_server.py	RoHaLocalFS bit-perfect file server (:8200). Range-aware (206/416), DLNA-headered, /localfs/stream/ + /localfs/art/. Path-traversal defended.
P/g12	dlna_localfs_wiring.py	Boot-time wiring of the LocalFs provider (maybe_start_localfs); gated on LOCALFS_MUSIC_ROOT / config.
P/g13	dlna_content.py	UPnP ContentDirectory SOAP client (cd_browse/cd_search). Reached only via P/g10 upnp.py.
P/g14	dlna_avtransport.py	UPnP AVTransport + RenderingControl SOAP (SetURI/Play/Stop/Pause, state/position probe, SetVolume).
P/g15	dlna_player.py	RendererQueue (sequential per-renderer playback, gapless via SetNextAVTransportURI, watchdog) + QueueRegistry (QUEUES, one per UDN).
P/g16	dlna_stream_proxy.py	Browser-audio HTTP proxy /stream (Range pass-through, 5-min idle timeout) + proxy_radio_stream /radio_stream (ICY de-interleave).
P/g17	dlna_art_fetcher.py	AlbumArtFetcher — MusicBrainz release-group + Cover Art Archive lookup; sticky notfound cache; ~1 req/s.
P/g18	dlna_acoustid.py	AcoustIDFetcher — fpcalc fingerprint → AcoustID → MusicBrainz metadata into metadata_overrides; transient-vs-permanent split; year backfill.
P/g19	dlna_lyrics.py	On-demand lrclib lyrics; cached in the lyrics table; sticky positive + negative.
P/g20	api_browse.py	Browse / search / radio-shuffle API.
P/g21	api_playback.py	Playback, /stream route, /art proxy, /api/client_log, state, indexer + AcoustID management.
P/g22	api_playlists.py	Playlist CRUD endpoints.
P/g23	api_upnp.py	UPnP service descriptors + ContentDirectory SOAP exposing the gateway's own playlists + favourites (so the Naim can browse it as a MediaServer).
P/g24	api_subsonic.py	Subsonic-compatible /rest/* API (browse, playlists, favourites→starred, stream, cover, scrobble, internet radio) for CarPlay clients.
P/g25	api_radio.py	Internet-radio endpoints: radio-browser search, favourites (≤25), ICY now-playing.
P/g26	dlna_config.py	Constants (DB_FILE/CFG_FILE/LOG_FILE), logging setup, config load/save, .env load (if dotenv present).

Maintenance tools (T/*) — tools/*.py

Operate directly on library.db and/or the music root. All default to dry-run / report-only; deletions move to Trash unless --hard-delete. Each has a unit-test sibling tools/test_*.py.

Code	File	Purpose & key options
T/a1	regen_schema.py	Regenerate the committed schema.sql artifact after any schema change. **--check** = fail if stale (the test_schema_sync gate).
T/a2	prune_empty_music_dirs.py	Trash directories whose subtree has zero music files. **--dry-run -v --limit N --exts a,b --hard-delete -y**.
T/a3	find_corrupt_audio.py	Flag files with bad/zero magic bytes for their extension. Writes corrupt-audio.txt. **--trash --hard-delete --limit --exts --out -y**.
T/a4	find_duplicate_audio.py	Find duplicate recordings on disk (same acoustid metadata); ranks a winner by bit-depth/sample-rate/size. **--trash --hard-delete -v -y**.
T/a5	retry_notfound_metadata.py	Drop bogus 'notfound' metadata_overrides so AcoustID retries. **--all | --since TS · --dry-run -y --log**.
T/a6	relink_orphan_overrides.py	Relink metadata_overrides orphaned by a UPnP rescan via d-id + fuzzy (artist,title). **--apply --db**.
T/a7	relink_playlists_to_localfs.py	Repoint playlist_tracks/favourites at RoHaLocalFS by normalised metadata; prune no-match. **--apply --no-backup --no-prune-favs -y**.
T/a8	audit_override_mismatches.py	Delete acoustid overrides whose artist AND title both mismatch the track (d-id collision repair). **--clean --top -y**.
T/a9	correct_year_drift.py	Rewrite metadata_overrides.year to the earliest plausible year evidenced by another copy in the library. **--apply --top --db -y**.
T/a10	improve_song_years.py	Query MusicBrainz for each song's earliest recording year → song_year_cache → apply. **--lookup --apply --limit -v**.
T/a11	localfs_scan.py	Standalone LocalFs indexer (CLI). NOTE: do not run against the live DB without a base_url — leaves placeholder URLs.
T/a12	localfs_serve.py	Standalone LocalFs file-server launcher for testing the :8200 serving path.
T/a13	beets_enrich.py	Run the beets tag-in-place enrichment batch (docs/enrichment.md). Safe wrapper around `beet import`: enforces write:yes/copy:no/move:no before any write. **--write-config --quiet --timid --album --revisit --reindex --gateway --dry-run -y**.

LAN devices (D/*), external services (E/*), scheduled jobs (J/*)

Code	Name	Role / protocol
D/1	Naim Uniti	UPnP MediaRenderer. Receives AVTransport+RenderingControl SOAP (control/volume); pulls audio bytes over HTTP Range from :8200; also browses gateway playlists as a control point. Gateway is NOT in its audio path → bit-perfect.
D/2	UPnP MediaServer	MinimServer / generic UPnP (AssetUPnP decommissioned). Browsed via P/g10 UpnpProvider + P/g13 ContentDirectory SOAP. Kept for any non-LocalFs source.
D/3	Music files	/Volumes/SAMDATA/Music (external drive, read-only). Indexed by P/g9; tags read by localfs provider; bytes served by P/g11.
D/4	fpcalc	Chromaprint CLI binary (brew). Local fingerprinter for P/g18. Internal/local — not network.
E/x1	musicbrainz.org	GET /ws/2/release-group — MBID + original year. UA + 1.1 s rate limit required.
E/x2	coverartarchive.org	HEAD /release-group/{mbid}/front-500 — cover presence.
E/x3	lrclib.net	GET /api/get — on-demand lyrics for the playing track.
E/x4	*.api.radio-browser.info	GET /json/stations/search — internet-radio catalogue (HLS filtered).
E/x5	api.acoustid.org	POST /v2/lookup — fingerprint → MusicBrainz metadata. Needs an APPLICATION key.
J/1	com.roha.dlna-gateway	LaunchAgent that runs the gateway. Restart: launchctl kickstart -k gui/$(id -u)/com.roha.dlna-gateway.
J/2	com.roha.dlna-cert-renew + renew-cert.sh	Weekly TLS cert renewal (Mon 04:30; no-op unless <30 days). cert-renewal.log.
J/3	com.roha.dlna-acoustid-retry + retry-acoustid-weekly.sh	Weekly: clear notfound + restart so AcoustID re-tries (Mon 05:30). acoustid-retry.log.
J/4	setup.sh	venv setup + run. --run --no-browser --debug --probe URL --list-devices --reset-devices.

Commands & options used in the drawing

<table>
<tr><th>Context</th><th>Command</th></tr>
<tr><td>Run (J/4)</td><td>./setup.sh --run [--no-browser] [--debug] [--probe http://…] [--list-devices] [--reset-devices]</td></tr>
<tr><td>Restart (J/1)</td><td>launchctl kickstart -k gui/$(id -u)/com.roha.dlna-gateway</td></tr>
<tr><td>Cert (J/2)</td><td>./renew-cert.sh [--force] · launchctl kickstart gui/$(id -u)/com.roha.dlna-cert-renew</td></tr>
<tr><td>AcoustID retry (J/3)</td><td>./retry-acoustid-weekly.sh [--dry-run]</td></tr>
<tr><td>Subsonic auth</td><td>launchctl setenv SUBSONIC_PASSWORD … ; launchctl getenv SUBSONIC_PASSWORD</td></tr>
<tr><td>Full test suite</td><td>python tests/run_all.py [--offline | --frontend | --frontend-only | http://host:8765]</td></tr>
<tr><td>Unit / Playwright</td><td>python3 -m unittest tests.test_player … ; .venv/bin/pytest tests/frontend -v</td></tr>
<tr><td>Chaos (live)</td><td>python3 tests/chaos.py --iterations N --workers M [--seed S] [--quiet] [--base https://host:8443]</td></tr>
<tr><td>Schema gate (T/a1)</td><td>python3 tools/regen_schema.py [--check]</td></tr>
<tr><td>Module self-tests</td><td>python dlna_discovery.py | dlna_content.py | dlna_library.py | db_pool.py | dlna_player.py | dlna_server.py</td></tr>
<tr><td>Range / 206 check</td><td>curl -r 0-1023 -D - http://<host>:8200/localfs/stream/<id> -o /dev/null</td></tr>
<tr><td>Tools (T/a2–a12)</td><td>python3 tools/<tool>.py [--dry-run|--apply|--trash|--hard-delete|--limit N|--exts a,b|-v|-y] (see the tools table above)</td></tr>
</table>

Ports: 8765 HTTP · 8443 HTTPS · 8200 RoHaLocalFS file server · 26125 (legacy AssetUPnP, decommissioned). See CLAUDE.md and REQUIREMENTS_2.0.md for the HTTP/2 + free-TLS roadmap.